

105 Green Revolution

107 A Store Lore

108 Remote IT Infrastructure Consolidation

GREEN REVOLUTION

Dr Kaladhar Voruganti

A datacentre that hosts application servers, network switches and storage devices is considered to be 'green' if it consumes less power for running and cooling the computer systems hosted by it. Datacentre operators are very conscious about power consumption because there is a limit on the amount of power that can be provided by the power sub-station supporting the datacentre. Once that limit is reached the datacentre operator has to open a new datacentre, and the new unit will initially not be fully utilised. Thus, in today's cost-conscious era, most datacentre operators are vigilant about ensuring that the computing resources are not under-utilised (because underutilised resources still consume a lot of power even when they are not doing useful work); procuring devices that consume less power and ensuring that they are using efficient datacentre cooling mechanisms. Thus, in this article we will discuss how datacentre administrators deal with the following issues:

There was a green revolution in the 1960s which increased the yield of agriculture production in many developing countries. similarly, there is another type of green revolution that is occurring in today's environmentally conscious world for it systems

- How to increase resource utilisation?
- How to evaluate the resources they are purchasing with respect to power?
- How to ensure the devices are consuming less power?
- How to design/leverage efficient datacentre cooling mechanisms?

Increasing resource utilisation

Memory, CPU, fan, network cards and disks are some important computer resources that consume power. The input workload determines how much work these resources have to perform, and this, in turn, dictates the amount of power consumed by the computer system. It is interesting to note that the amount of power consumed by a system that is 50 per cent utilised is not that much more than a computer system that is 10 per cent utilised (due to the presence of fixed power consumption costs). Hence, less power is consumed by a single box that is 50 per cent utilised than five boxes that are each only 10 per cent utilised.